

FLAPPY JAVA

Endless Flyer

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Review Number: 1

GAME CONCEPT OVERVIEW



Core Mechanics

A "one-tap" side-scrolling survival game where the player controls a bird navigating through narrow gaps.



Addictive Loop

Simple tap to jump against constant downward gravity. The loop is short, high-stakes, and infinitely repeatable.



Win/Loss Condition

Goal: Pass as many pipes as possible. **Lose:** Hit a pipe, the ground, or the sky ceiling.

REFERENCE GAME STUDY

Analysis of Flappy Bird

Original success relied on "**Punishing Simplicity**"—easy to understand but extremely hard to master.

- **Reproduced:** Gravity-velocity physics, collision hitboxes, and random pipe heights.
- **Simplified:** Use of Java Graphics2D instead of a heavy physics engine; AABB (Rectangular) collision logic.



TARGETED LEARNING OUTCOMES

 **Multithreaded Real-time Loops:** Implementing a consistent 60FPS refresh rate without blocking the UI.

 **Collision Detection Logic:** Real-time overlap calculation between Bird and Pipe entities.

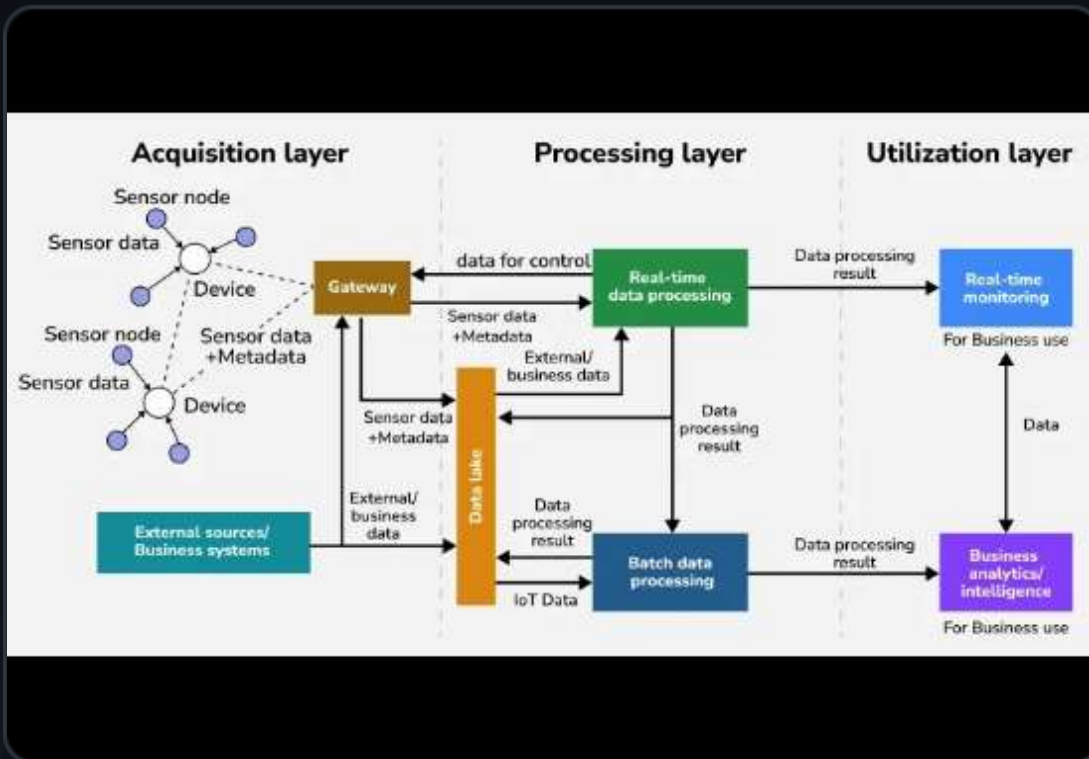
 **Custom 2D Rendering:** Direct manipulation of Java AWT Graphics2D for high-performance redraws.

 **JDBC Persistence:** Managing a persistent leaderboard using SQL connectivity for high scores.

PROPOSED MODULE BREAKDOWN

Module	Core Functionality	Java Feature Applied
Game Loop	Continuous thread for physics and redraws	Multithreading (Runnable)
Collision Engine	Boundary & overlap checking logic	AABB Logic / Rectangle.intersects()
Swing Rendering	Key listeners and sprite drawing	Graphics2D / Event Handling
Score & DB	Leaderboard persistence and retrieval	JDBC / MySQL / Collections
Difficulty Scaling	Dynamic speed and gap adjustment	Algorithms / State Mgmt

SYSTEM ARCHITECTURE



Data Flow Interaction

- 1. Input:** Keyboard events trigger velocity changes in the Bird Model.
- 2. Game Loop:** Background thread updates coordinates and checks Collision Engine.
- 3. Renderer:** Redraws updated models using Double Buffering.
- 4. DB Layer:** Writes score to MySQL on the "Game Over" state.

TECH STACK & WHY



Swing/AWT

Used for building the game window and utilizing the robust Graphics2D library for sprites.



ArrayList

Chosen for the **Collections Framework** to manage an infinite stream of dynamic pipe objects.



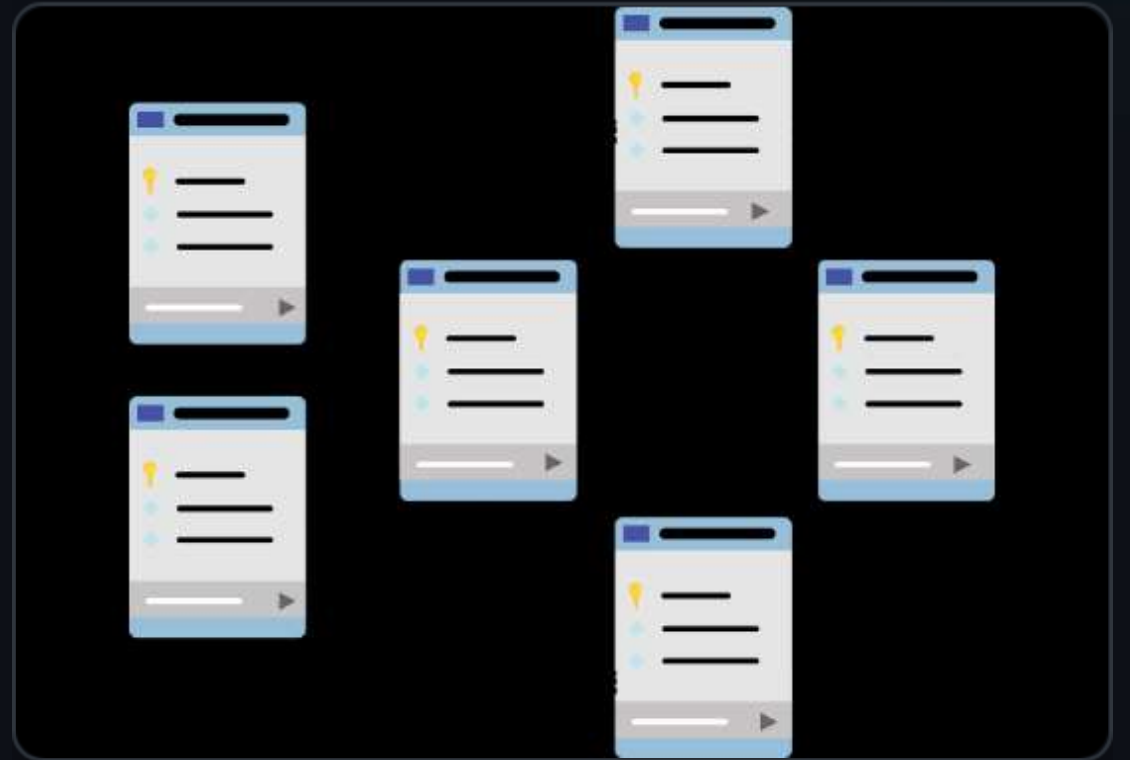
MySQL + JDBC

Provides permanent storage for players' names and high scores beyond individual game sessions.

PROPOSED DATABASE SCHEMA

```
CREATE TABLE leaderboard (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  player_name VARCHAR(50),  
  pipes_passed INT,  
  played_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

The table will record the score at the exact moment of collision (Game Over).



MULTITHREADING PLAN

01

Dedicated Game Thread

The entire game logic (physics, movement, collisions) will run on a separate Thread. This prevents the **GUI Freeze** that occurs if long-running loops or `Thread.sleep()` were executed on the Event Dispatch Thread (EDT).

"Separating Logic from Presentation ensures smooth animation while remaining responsive to user keyboard inputs."

EXCEPTION HANDLING PLAN

SQL Connectivity

SQLException: Handled with try-catch blocks to ensure the game doesn't crash if the database is offline.

Asset Loading

IOException: Gracefully manages missing bird or pipe sprites by reverting to colored rectangles if files fail to load.

Thread Interrupts

InterruptedException: Safely shuts down the game loop thread during program exit or window closing.

WORK PLAN & TIMELINE



CHALLENGES & Q&A

- ⚠ **Hitbox Precision:** Handling collision between rectangular pipes and non-rectangular bird shapes.
- 🎮 **Frame Rate Sync:** Keeping animation smooth across different hardware performance levels.
- 🔗 **Thread Safety:** Safely updating shared state variables between the logic thread and drawing thread.



IMAGE SOURCES



https://market-resized.envatousercontent.com/previews/files/181055324/flappy-winter-bird-game-character-sprite-sheet-sidescroller-game-asset-animation-gui-mobile-games-gameart.jpg?w=590&h=590&cf_fit=crop&crop=top&format=auto&q=85&s=6d1065bca2cda06338bf71f911faf83d05d9a3616b42164428f0772c7d3e5376

Source: [graphicriver.net](https://www.graphicriver.net)



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Source: www.vecteezy.com